```
classDiagram

User <|-- Admin
IRequester <|.. User
IProvider <|.. User

IRequester "1" --> "*" Task : 发布
IProvider "1" --> "*" Order : 接单

Category "1" <-- "*" Task : 属于

Task "1" *-- "0..*" Order : 包含
Order "1" *-- "0..1" Review : 包含

User "1" <-- "*" Notification : 接收

class IRequester{
    <<interface>>
    +publishTask(Task task) void
    +cancelTask(Task task) void
}

class IProvider{
    <<interface>>
    +acceptTask(Task task) Order
}

class User{
    -Long id
    -String name
    -String email
    -String phone
    -String passwordHash
    -BigDecimal creditScore
    -Integer status

    +login() Boolean
    +logout() void
}

class Admin{
    -String adminLevel

    +handleReport(Report report) void
```

```
        +auditContent() void
        +banUser(User user,int days) void
}

class Category{
        -Integer id
        -String name
        -String icon
        -Boolean hasReview
        -Integer sortOrder

        +isReviewAllowed() Boolean
}

class Task{
        -Long id
        -IRequester requester
        -Category category
        -String title
        -String description
        -BigDecimal reward
        -TaskStatus status
        -String location
        -LocalDateTime deadline
        -LocalDateTime publishedAt

        +validateDeadline() Boolean
        +updateStatus(TaskStatus newStatus) void
}

class Order{
        -Long id
        -Task task
        -IRequester requester
        -IProvider provider
        -OrderStatus status
        -LocalDateTime acceptedAt
        -LocalDateTime completedAt
        -LocalDateTime cancelledAt

        +completeOrder() void
        +cancelOrder() void
}
```

```
class Review{
        -Long id
        -Order order
        -Integer score
        -String comment
        -LocalDateTime createdAt

        +submitReview() void
        +calculateCreditImpact() Integer
}

class Notification{
        -Long id
        -User receiver
        -String type
        -String title
        -String content
        -Boolean isRead

        +markAsRead() void
}
class Report{
}

class TaskStatus{
        <<enumeration>>
}

class OrderStatus{
        <<enumeration>>
}
```


«interface»
IRequester
+publishTask(Task task) : void
+cancelTask(Task task) : void
发布
Category
-Integer id
-String name
-String icon
-Boolean hasReview
-Integer sortOrder
+isReviewAllowed() : Boolean
Report
«enumeration»
TaskStatus
«enumeration»
OrderStatus
属于
Task
-Long id
-IRequester requester
-Category category
-String title
-String description
-BigDecimal reward
-TaskStatus status
-String location
-LocalDateTime deadline
-LocalDateTime publishedAt
+validateDeadline() : Boolean
+updateStatus(TaskStatus newStatus) : void
«interface»
IProvider
+acceptTask(Task task) : Order
接单
包含
User
-Long id
-String name
-String email
-String phone
-String passwordHash
-BigDecimal creditScore
-Integer status
+login() : Boolean
+logout() : void
Order
-Long id
-Task task
-IRequester requester
-IProvider provider
-OrderStatus status
-LocalDateTime acceptedAt
-LocalDateTime completedAt
-LocalDateTime cancelledAt
+completeOrder() : void
+cancelOrder() : void
接收
包含
Admin
-String adminLevel
+handleReport(Report report) : void
+auditContent() : void
+banUser(User user,int days) : void
Notification
-Long id
-User receiver
-String type
-String title
-String content
-Boolean isRead
+markAsRead() : void
Review
-Long id
-Order order
-Integer score
-String comment
-LocalDateTime createdAt
+submitReview() : void
+calculateCreditImpact() : Integer